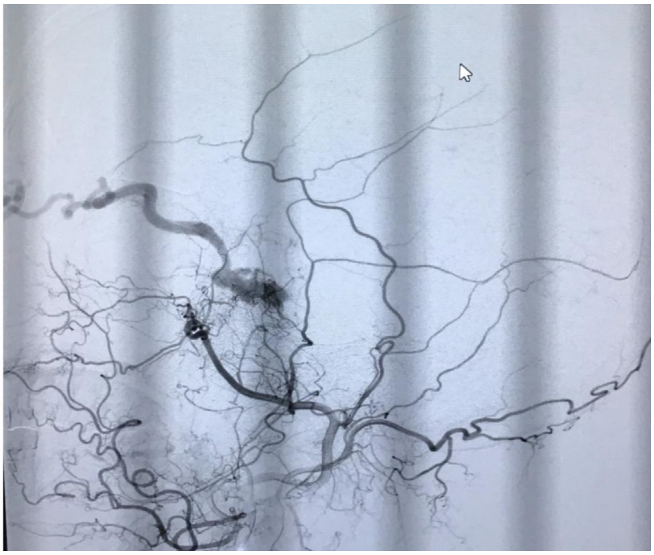


Methods

45-year-old woman with no history of trauma presented with progressive right eye redness, swelling, and pulsatile tinnitus for three months. Examination revealed right-sided proptosis, conjunctival congestion, chemosis, and a palpable thrill with an audible bruit over the periorbital region. Mild cranial nerve VI palsy was also noted. Mri Brain suggested abnormal cavernous sinus flow, and digital subtraction angiography (DSA) confirmed an indirect CCF. Endovascular embolization was planned. **Intervention** The patient underwent transvenous embolization via the inferior petrosal sinus. Detachable coils were deployed to occlude the arteriovenous shunt. Post-procedure angiography confirmed complete obliteration of the fistula.

Results

Post-procedure, the patient reported resolution of bruit and significant reduction in proptosis within one week. Follow-up imaging at three months showed no residual fistula or venous congestion. Neurological examination revealed complete recovery of ocular movements.

Conclusions

This case emphasizes the necessity of early recognition of spontaneous CCFs and highlights the efficacy of transvenous embolization. Awareness among clinicians can facilitate timely referral and intervention, preventing long-term morbidity associated with delayed treatment.

doi:10.1016/j.jns.2025.125429

125430
VIRTUAL-0370 / #3597

Topic: *AS34. Neuroophthalmology*

A case of posterior ischemic optic neuropathy as a rare cause of bilateral vision loss

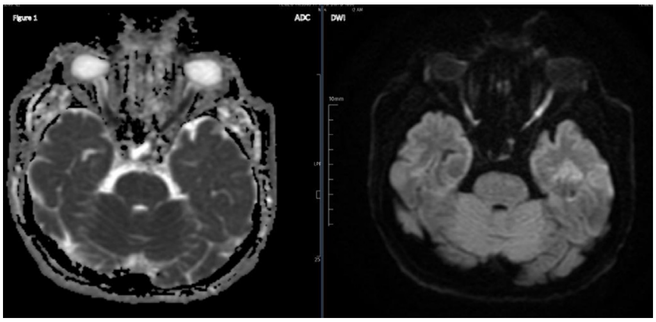
Funda Uysal Tan[a], Melis Doğanay Öcalan[a], Saniye Üke Uzun[b], *[a]Etlik Cıty Hospital, Neurology, Ankara, Turkey, [b]Etlik Cıty Hospital, Ophthalmology, Ankara, Turkey*

Background and aims

Posterior Ischemic Optic Neuropathy (PION), a relatively rare condition, is primarily identified by the sudden onset of visual impairment, characteristic optic nerve-related visual field defects, and an initially normal optic disc reflecting its acute ischemic pathology. Although PION is generally a diagnosis of exclusion, the introduction of modern magnetic resonance imagıng (MRI) techniques such as diffusion-weighted imaging (DWI) and apparent diffusion coefficient (ADC) mapping offers stronger evidence for precise diagnosis and management.

Methods

A 60-year-old man with a history of hypertension presented with sudden onset of blurred vision followed by vision loss. Within one day, a comparable pathology is observed in the right eye as well. The best visual acuity was no light perception with negative projection in the left eye and finger counting in the right eye, however, no optic disc edema was detected in the fundus of either eye. Laboratory and cerebrospinal fluid examinations were normal. However, his brain MRI revealed a focal high hyperintensity signal at the left optic nerve on the T2 DWI series. The area was corresponded with the hypointensity area in the ADC series (Fig. 1), which was a powerful clue for PION.



Results

At 3 weeks after the visual loss, an examination revealed stationary visual acuity in both eyes, bilateral pale optic discs, and ganglion cell loss in both eyes.

Conclusions

PION is usually diagnosed by excluding other conditions; however, careful analysis of the DWI and ADC sequences on MRI can provide clinicians with stronger evidence for a definitive diagnosis and guide appropriate treatment.

doi:10.1016/j.jns.2025.125430

125431
VIRTUAL-0371 / #1349

Topic: *AS35. Neurootology - Vestibular disorders*

Pulsatile neurovascular compression: Vertigo and hemifacial spasms from vertebral artery and PICA conflict with potential flow-dynamic correlates

Halwan Bayuangga, Cempaka Setyaningrum, *Gadjah Mada University, Department of Neurology, Yogyakarta, Indonesia*

Background and aims

Neurovascular compression syndromes (NVC) occur when blood vessels compress cranial nerves, causing dysfunction. While hemifacial spasm (HFS) typically result from facial nerve (CN VII) compression and vertigo from vestibulocochlear nerve (CN VIII) involvement, concurrent compression of both nerves is rare. This case highlights the diagnostic challenges and management considerations in dual NVC, emphasizing the potential role of vascular tortuosity in symptom pathogenesis.

Methods

Case report.

Results

A 60-year-old female presented with a 4-year history of recurrent, positionally exacerbated vertigo episodes and progressive left HFS characterized by intermittent periocular-perioral twitching. The patient reported occasional ipsilateral pulsatile tinnitus but denied hearing loss. Neurological examination revealed triggered left HFS during voluntary movements, with preserved motor/sensory function. The Dix-Hallpike maneuver was negative for BPPV. Neuroimaging revealed a tortuous left vertebral artery compressing CN VII at the root entry zone and a dolichoectatic PICA loop compressing CN VIII at the cerebellopontine angle. The tortuosity-induced hemodynamic stress may exacerbate pulsatile neurovascular compression and drive episodic symptoms (vertigo/spasms), worsened by exertion. No bridging vein compression was seen on MRI, ruling out venous congestion as a contributing factor. Medical management with Carbamazepine 200 mg BID and Betahistine 24 mg BID resulted in partial symptomatic relief of HFS and vertigo, respectively.

Conclusions

This case demonstrates that dual NVC require comprehensive evaluation combining high-resolution neuroimaging and functional assessment. The tortuous vertebrobasilar anatomy may contribute to both structural compression and potential hemodynamic effects. While pharmacotherapy provided partial resolution, microvascular decompression success may be dependent on both careful mechanical dissection and flow-dynamic modulation.

doi:10.1016/j.jns.2025.125431

125432
VIRTUAL-0372 / #1351

Topic: *AS35. Neurootology - Vestibular disorders*

The Syncope-vertigo paradox: Vestibular paroxysmia as a stealth cause of recurrent transient loss of consciousness

Cempaka Setyaningrum, Halwan Bayuangga, *Gadjah Mada University, Department of Neurology, Yogyakarta, Indonesia*

Background and aims

Vestibular paroxysmia (VP) is a rare but treatable neurovascular disorder characterized by brief, recurrent attacks of vertigo caused by compression of the vestibulocochlear nerve (CN VIII) by adjacent vessels, most commonly by the anterior inferior cerebellar artery (AICA). We report challenges in diagnosing VP, particularly when vertigo is overshadowed by syncope or autonomic symptoms.

Methods

Case report.

Results

A 28-year-old female with a medical history of cerebral small vessel disease (CSVD) presented with recurrent episodes of sudden dizziness followed by syncope occurring 2–3 times weekly for 2 months. The episodes were characterized by brief (5–15 s) intense vertigo described as a “spinning sensation” immediately preceding loss of consciousness, without identifiable triggers or associated symptoms. Physical examinations revealed mild hyperreflexia (likely related to CSVD) but was otherwise unremarkable. Neuroimaging revealed neurovascular compression of the left CN VIII by AICA, right vertebral artery hypoplasia, and a non-visualised left transverse sinus, alongside Fazekas grade 1 white matter changes. Cardiac and vestibular testing were unremarkable. Oxcarbazepine 150 mg OD resulted in significant reduction in episode frequency with a dose escalation plan to 300 mg BID to achieve complete symptom resolution at 6–12 month follow up.

Conclusions

This case highlights vestibular paroxysmia as a differential in recurrent syncope when cardiac and epileptic causes are excluded. The patient's CSVD, vertebral artery hypoplasia, and transverse sinus anomalies underscore the importance of comprehensive neuroimaging in unexplained vertigo/syncope cases. Her favourable response to